Perishable Demand Forecasting & Zero-Waste Inventory Engine

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Introduction

A stockout records zero sales, not zero demand. A forecast trained on raw sales keeps under-ordering exactly the products that sell out.

Objective: Recover the demand that stockouts hide, forecast a realistic best-to-worst-case range for it, and use that to decide exactly how much stock to order: a better order, not just a better forecast.

Success: cheaper than *"order what sold last week"* at every plausible cost ratio.

Data

Subset of FreshRetailNet-50K: 100 stores, 5,601 series, 43.8% of product-days stocked out (Partially).

- Dingdong Inc. (grocery chain), 100 of 898 stores picked at random (about 11% of the full dataset).
- Four blocks in time order, never overlapping: 62 days to train → **validation weeks** (2 weeks, to compare models and pick the best one) → **calibration weeks** (2 weeks, to adjust the forecast ranges) → **sealed test week** (opened once, at the end).
- What makes this solvable: the data records which hours each shelf was empty, so "nothing sold" and "nothing left to sell" look different.

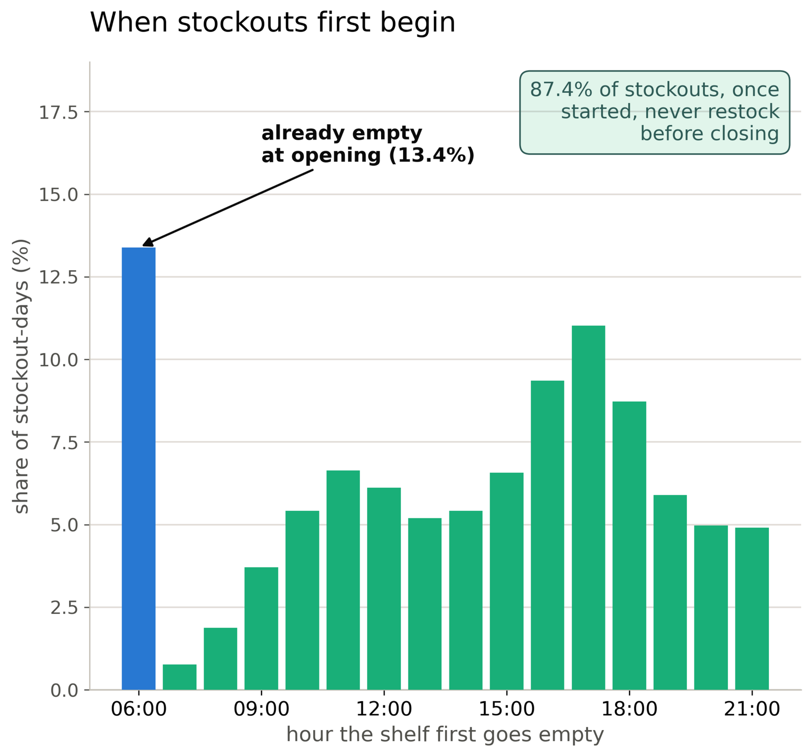

Figure 1. *Stockout timing.*

Once out, it stays out: 87.4% never restock before closing (13.4% already empty at opening).

Approach & Workflow

Ingest & Subset
seeded draw
A representative, whole-store draw: 100 seeded stores, 5,601 series, frozen calendar.

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Recover
LightGBM → fill + correct
Raw sales undercount once a shelf empties. Recovery fills those hours with a modelled selling rate, floored at what was observed.

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Forecast
TFT + XGBoost, 6 quantiles
Two model types × two data types = 4 versions, scored on validation weeks never trained on. Six quantiles each, not just the median.

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Calibrate
split-CQR (split conformal quantile regression)
Widens the forecast range so a stated 80% band actually contains truth 80% of the time.

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Order & Evaluate
newsvendor rule · cost sweep
Converts the forecast into an order at the percentile a cost ratio implies, checked against naive practice under 9 assumptions.

Stage-wise Results

Each stage checked on its own terms before the next was built on it.

Table 1. *Result checked at each pipeline stage.*

Recovering hidden demand vs. no-model average **0.30 vs. 0.37 WAPE** (19% better)	**Forecast accuracy, chronic-stockout products** ≥75% censored series only, not the pooled average **WAPE falls 25.2%** (TFT), **22.9%** (XGBoost), **vs. raw**
Calibrated forecast ranges 80% band's actual coverage (TFT, recovered) **82.6%** (validation), **79.1%** (test); **target 80%**	**Waste at 95% demand met** TFT: raw vs. recovered **44.0%→41.4%** (validation); **72.7%→51.9%** (test)

Final Ordering Decision

Order the q^*-th percentile of the calibrated, recovered TFT forecast.

The newsvendor rule is the classic formula for the order quantity that minimises cost when a stockout and a wasted unit cost different amounts. The optimal order = $F^{-1}(q^*)$, $q^* = \frac{c_u}{c_u+c_o}$. c_u: cost of running short by one unit. c_o: cost of one unit left over.

Tested across nine cost assumptions in total, from a stockout costing a quarter as much as a wasted item to 19 times as much. The model beat today's practice at every cost ratio.

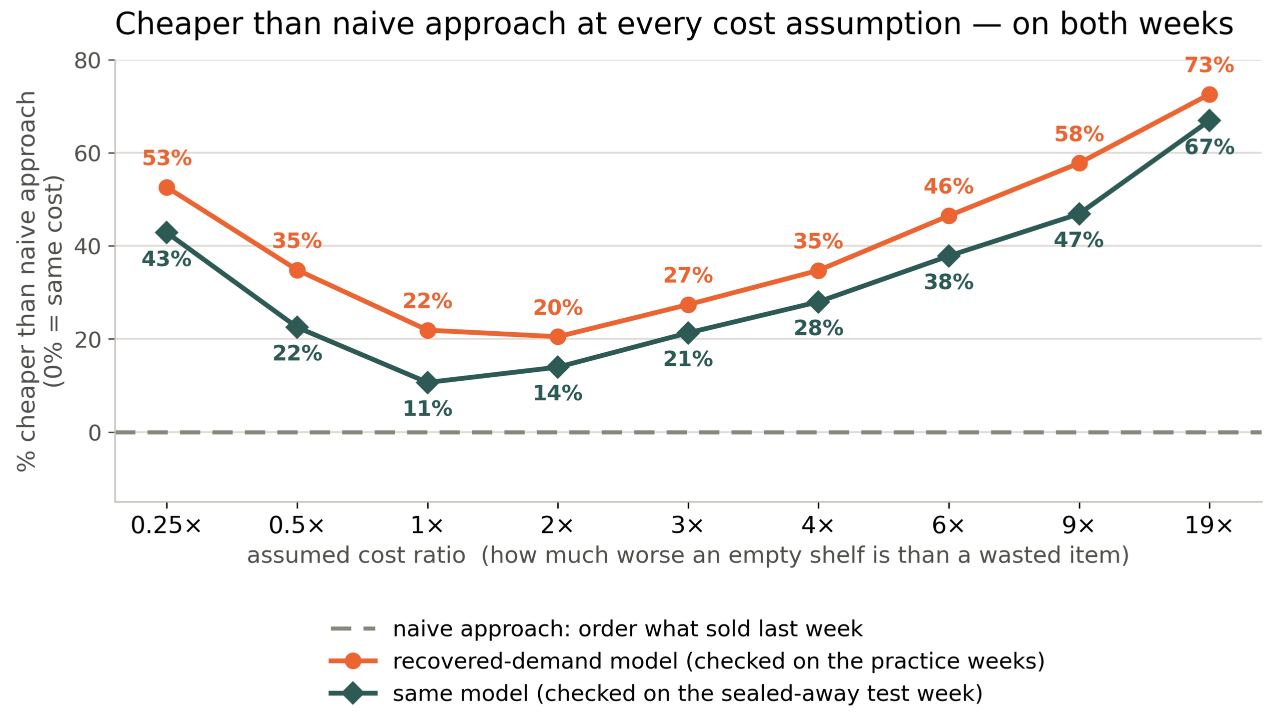

Figure 2. *Cost sweep vs. naive rule.*

The savings are smaller on the sealed test week than on the validation weeks, but still real at every cost assumption. Both are averages over a handful of days, so treat them as ranges, not exact figures: the true saving is likely 30–39% on the validation weeks and 25–31% on the test week.

Choosing an Operating Point

No single policy wins outright - the choice is which risk to carry.

The three policies below are the same recovered-demand forecast, each ordering at a different percentile. The **naive rule** is simulated too, not raw data: it orders whatever sold last week.

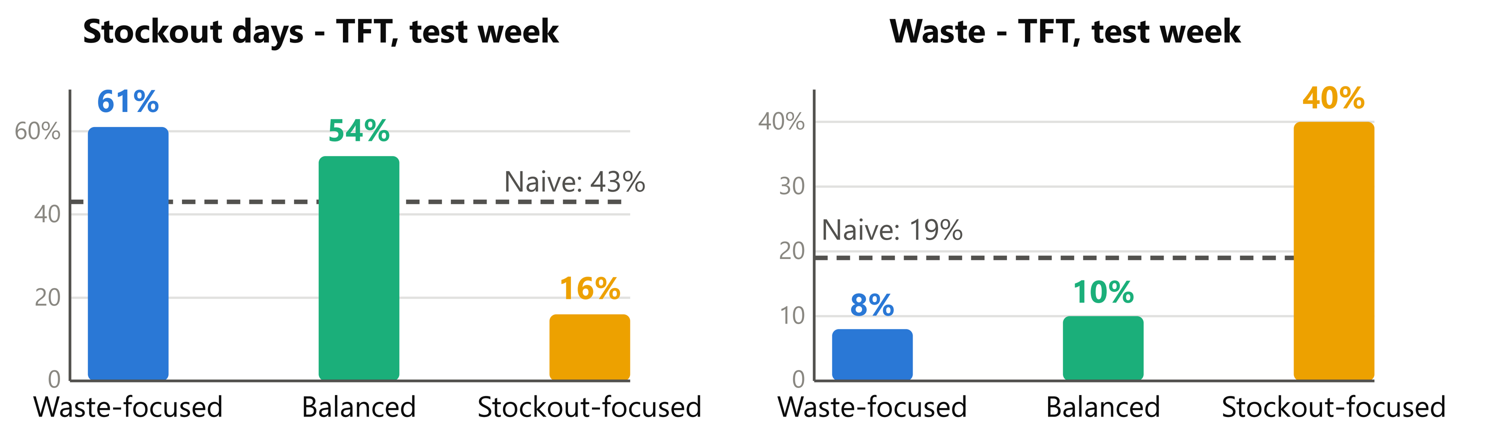

Figure 3. *Policy comparison, TFT, test week.*

Test week only: the sealed, honest evaluation. Dashed line = naive-approach; bars below it beat naive, bars above it don't. Validation figures are in the tag below.

Balanced cuts test-week waste to 10% (naive: 19%) but stockout days rise to 54% (naive: 43%) - the reverse of validation, where Balanced beat naive on both (35% stockout days, same waste). Why the reversal: test-week demand ran ~36% higher than training. Stockout-focused is the only policy beating naive on stockouts in both periods, at far more waste.

Same reversal on XGBoost. All three stay cheaper than naive on cost at every ratio tested (Final Ordering Decision) - the choice here is which risk to carry, not which saves money.

One Store, Test Week

Store #76 (closest to median in the subset: 43.8% of product-days out of stock, 57 products), at a stockout costing 2× a wasted item: stockout days **34.7%** (of product-days); demand met **90.5%**, waste **17.1%** (both % of demand); cheaper than naive **26.6%** (vs. last week's sales).

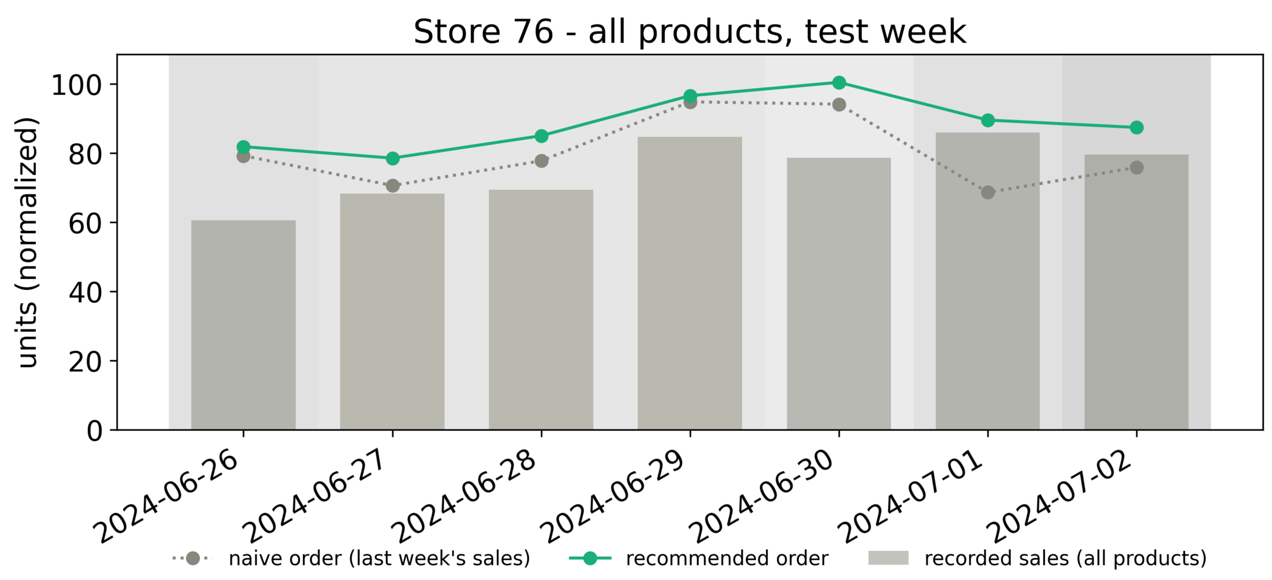

Figure 4. *One store, test week.*

Illustrative only. The 100-store aggregate is the evidence. Darker shading = more of the store was stocked-out.

Recovery vs. Model Choice

All four versions (TFT/XGBoost × raw/recovered) were ordered and scored the same way against the naive rule, at the cost assumption $c_u/c_o=4$ (a stockout costs 4× a wasted item), on the sealed test week. Higher is better.

Table 2. *Percent cheaper than naive, sealed test week, by architecture and training target.*

% cheaper than naive	Test week
TFT, raw	9.2%
TFT, recovered	28.0%
XGBoost, raw	0.8%
XGBoost, recovered	13.4%

- **Recovery's Effect:** fixes censored products specifically - proven on validation (Stage-wise Results: WAPE falls sharply in the chronic band). Test-week demand ran **~36% higher**, so the fix paid off everywhere: up to **+18.8 pts** cheaper than naive.
- **Model Choice's Effect:** real, but smaller - TFT beats XGBoost (**+14.6 pts** cheaper than naive). TFT is the headline model, the more accurate forecast on validation (pinball 0.1074 vs. 0.1116).

Bottom line: correct the data before you trust the forecast.

Novelty

- **Not just a new model:** a two-stage LightGBM correction turns censored demand into a cleaner target; testing TFT and XGBoost on both raw and corrected data isolates architecture from that fix.
- **All the way to an order:** the pipeline runs to an order quantity (newsvendor, split-CQR calibrated), judged on cost, waste and stockouts, not accuracy alone.

Limitations & Future Work

A proof of concept on a dataset that records sales, not stock.

Table 3. *Known limitations and what they mean.*

No true demand on stockout days Checked on full-shelf days instead; no stock-on-hand data to confirm the worst days directly.	**One store sample, one calendar** One random draw of 100 stores. We tested how much savings could wobble, not a different draw's winner.
Cost ratio is assumed Tested across a range (Final Ordering Decision), not measured.	**Future work** Stock-on-hand/delivery data would confirm recovery directly; recorded spoilage would make waste observed.

Conclusion

Fixing the data mattered more than the model.

A stockout corrupts the training label, not just the sale (previous card). A better architecture can't undo a bias baked into the data; only recovering it can.

Bottom line: every policy tested is cheaper than *"order what sold last week"*. Which one to run is a store's call, based on whether stockouts or waste cost it more.

References

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- Lim et al. (2021). Temporal Fusion Transformers for interpretable multi-horizon forecasting.
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